

# Ordinary Differential Equations (ODE)

## Ordinary Differential Equations (ODE):

Equalities involving unknown (1-variable) functions and their derivatives.

(Partial Differential Equations (PDE):

unknown (multi-variable) functions and their partial derivatives.)

Order of an ODE: the highest order of derivatives.

$n^{\text{th}}$ -order ODE:  $F(y^{(n)}, \dots, y', y, x) = 0$ , or in the form

$$y^{(n)} = f(y^{(n-1)}, \dots, y, x).$$

$y$ : the unknown function;  $x$ : the independent variable.

**Example 1** A body of mass  $m$  falls under gravity  $g$  with a drag force proportional to the square of the velocity (with constant of proportionality  $\mu$ ).

The downward velocity  $v(t)$ :  $m \frac{dv}{dt} = mg - \mu v^2$ .

**Example 2** Chemical reaction:  $A + B \rightarrow C$ .

Their concentrations  $a$ ,  $b$  and  $c$  depending on time  $t$ :

$$\frac{da}{dt} = -kab, \quad \frac{db}{dt} = -kab, \quad \frac{dc}{dt} = kab,$$

$k$  rate constant. (A system of ODEs).

- Different Types of ODEs.
- Methods of solving or analysing them.

(Not all ODEs can be 'solved'.)

# First-Order ODEs

$$y' = f(x, y)$$

**Example**  $y' = x$

## Directly Integrable Equations $y' = f(x)$

- In general, solutions of  $y' = f(x, y)$ : not unique.

Initial Value Problem  $y' = f(x, y), y(x_0) = y_0$ : unique.

(Various sufficient conditions exist. E.g., Thm in next slide)

(Not always true. E.g.,  $y' = 2\sqrt{y}$ ,  $y(0) = 0$ : not unique!)

### Theorem

*$f$  and  $\frac{\partial f}{\partial y}$  are continuous in some rectangle  $\alpha < t < \beta$ ,  $\gamma < y < \delta$  containing  $(t_0, y_0)$ .*

*Then, in some interval  $t \in (t_0 - h, t_0 + h) \subset (\alpha, \beta)$ , there is a unique solution  $y = \phi(t)$  of the initial value problem*

$$y' = f(t, y), \quad y(t_0) = y_0.$$

**Proof** by Picard's iteration method: omitted.

$$\phi_0(t) = 0, \quad \phi_{n+1}(t) = \int_{t_0}^t f(s, \phi_n(s)) ds \rightarrow \phi(t)$$

**Example**  $y' = xy$

## Separable Equations $y' = f(x)g(y)$

- The solution may be of no explicit form. E.g.,  $\frac{dy}{dx} = \frac{x^2+1}{\cos y+1}$ .



**Example**  $y' = x + y$

**Linear Equations**  $y' + p(x)y = f(x)$

$f = 0$ : homogeneous;  $f \neq 0$ : inhomogeneous.

E.g.,  $y' + xy = e^x$ ,  $xy' + y = x^2$ , etc.

( $\mathcal{L} = \frac{d}{dx} + p(x)$  a linear differential operator:  $\mathcal{L}(y) = \frac{d}{dx}y + p(x)y$  linear.)

Two methods:

- Integrating factor ( $\mu(x) = e^{\int p(x)dx}$ )
- Variation of constant (Homogeneous  $\rightarrow$  Inhomogeneous)

**Example**  $2x + y^2 + 2xyy' = 0.$

**Exact Differential Equations**  $M(x, y) + N(x, y)y' = 0$   
where  $M_y = N_x$

- $M(x, y)dx + N(x, y)dy$ : exact.

**Example**  $(3xy + y^2) + (x^2 + xy)y' = 0.$

**Integrating Factor**  $\mu(x, y)$ :  $M(x, y) + N(x, y)y' = 0$

- $\mu(x, y)$  satisfies a PDE; may be difficult to find.
- Cases when  $\mu$  can be found:

$$\frac{M_y - N_x}{N} \text{ is a function of } x \text{ only} \rightarrow \mu = \mu(x) \checkmark$$

$$\frac{M_y - N_x}{M} \text{ is a function of } y \text{ only} \rightarrow \mu = \mu(y) \checkmark$$

- A special case: integrating factor for 1<sup>st</sup>-order linear ODE.

**Example**  $\frac{dy}{dx} = \frac{x^2+y^2}{xy}, \quad \frac{dy}{dx} + xy = x^3y^2.$

## Substitution Method (can be difficult, if exists)

- Homogeneous DE  $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$ :  $u = \frac{y}{x}$ .
- Bernoulli DE  $y' + p(x)y = q(x)y^n$ :  $u = y^{1-n}$ .
- Other cases: e.g.,  
 $f(x - y + c)y' - f(x - y + c) + g(x) = 0,$   
 $xy\frac{dy}{dx} + (x^2 + y^2 + x) = 0,$   
 $(\ln y - x)\frac{dy}{dx} - y \ln y = 0, \dots$

**Example**  $y' = \frac{y}{x-y^2}.$

**Other Methods**  $x(y)$ :  $y$  as independent variable;

...

try computing software; ...

- Recently, an AI beats Mathematica in solving DE.
- Unfortunately, many 1<sup>st</sup>-order ODEs cannot be solved analytically.

**Example**  $y' = x + y^3$ .

Visual Approximation: Slope Field

(Derivative  $\leftrightarrow$  Slope)

slope field  $x + y^3$

Numerical Approximation: Discretization

(Derivative  $\leftrightarrow$  Difference Quotient)

E.g.,  $y(x_0) = y_0$ ,

$$y(x_{n+1}) \approx y(x_n) + y'(x_n)(x_{n+1} - x_n) = y(x_n) + f(x_n, y(x_n))(x_{n+1} - x_n)$$

## A Summary of 1<sup>st</sup>-Order ODEs

- Directly Integrable, Separable, Linear
- Exact & Integrating Factor
- Substitution Method: Homogeneous, Bernoulli, etc.
- Reverse Independent and Dependent Variables, Computing Software.
- Visual Analysis, Numerical Approximation.
- . . . . .

## Second-Order Linear ODEs

$$y'' + p(x)y' + q(x)y = f(x)$$

$$(L(y) = f, L = \frac{d^2}{dx^2} + p(x)\frac{d}{dx} + q(x))$$

(Higher-Order: Similar)



# Why Linear?

$$L(y) = y'' + p(x)y' + q(x)y = f(x):$$

$f = 0$ : homogeneous or unforced;

$f \neq 0$ : inhomogeneous or forced.

## • Principle of Superposition

Homogeneous equation  $Ly = 0$ :

$\{y | Ly = 0\}$  is a vector space.  $\dim = 2$ .

( $\Leftarrow$  uniqueness of  $Ly = 0, y(x_0) = a, y'(x_0) = b$ .)

Inhomogeneous equation  $Ly = f$ :  $y = y_p + y_c$ ,

$y_p$ : a particular soln (of  $Ly = f$ );

$y_c$ : complementary function (general soln of  $Ly = 0$ .)

## Theorem

*Consider an initial value problem*

$y'' + p(x)y' + q(x)y = f(x)$ ,  $y(x_0) = y_0$ ,  $y'(x_0) = y'_0$ .  
 $p, q, f$ : continuous on an open interval  $I$  containing  $x_0$ .  
*Then, the initial value problem has exactly one solution, and the solution exist throughout the interval  $I$ .*

Proof: omitted.

Consider an example  $y'' = x$ .

IVP: 1-dim motion with initial position and velocity.

• **Wronskian**  $W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}.$

(Similar for  $n^{\text{th}}$ -order linear ODE.)

•  $W(y_1, y_2) \neq 0 \Rightarrow y_1, y_2$  lin. indep.

$W(y_1, y_2) = 0$  &  $Ly_1 = Ly_2 = 0 \Rightarrow$  lin. dep.

( $\Leftarrow$  uniqueness of initial value problem.)

(“ $W = 0$  at one point  $\Rightarrow W \equiv 0$ ” also follows.)

- **Wronskian**  $W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}.$

(Similar for  $n^{\text{th}}$ -order linear ODE.)

- $Ly_1 = Ly_2 = 0 \Rightarrow W(x) = e^{-\int p dx}$  ( $\Leftarrow W' = -pW$ )  
 ( $W$ : same for any two solns up to a multi. const.  
 ( $\dim\{y : Ly = 0\} = 2$ );  
 intrinsic of the ODE (application to come);  
 $W = 0$  at one point  $\Rightarrow W \equiv 0$ .)

## Second-order Linear ODEs: Constant Coefficients

**Example**  $y'' - 2y' - 3y = 0$ .

- $L = \frac{d^2}{dx^2} - 2\frac{d}{dx} - 3 = (\frac{d}{dx} - 3)(\frac{d}{dx} + 1)$ .
- Try  $y = e^{rx}$ . Characteristic equation.

**Example**  $y'' - 2y' + 2y = 0$ .

- Try  $y = e^{rx}$ . Complex-valued  $\leftrightarrow$  Real-valued.

**Example**  $y'' - 2y' + y = 0$ .

- Try  $y = e^{rx}$ . Repeated root  $\rightarrow$  verify  $y = xe^x$ .
- Why the form  $y = xe^x$ ?
  - $L = \left(\frac{d}{dx} - 1\right)^2$ .
  - Distinct  $\rightarrow$  repeated:  $\frac{e^{r_2x} - e^{r_1x}}{r_2 - r_1}$ ,  $r_2 \rightarrow r_1$ .
  - $L(e^{rx}) = (r - 1)^2 e^{rx}$ . Consider  $\frac{\partial}{\partial r} L(e^{rx})$ .
  - Use Wronskian  $W = e^{2x} = y_1 y_2' - y_1' y_2$ .
  - Reduction of order: try  $y(x) = v(x)y_1(x)$

(The latter two: applicable in general cases.)

## 2<sup>nd</sup>-Order Linear ODE w. Const. Coeff., Homogeneous

$L = a \frac{d^2}{dx^2} + b \frac{d}{dx} + c$ , characteristic equation  $ar^2 + br + c = 0$ :

- distinct roots;
- complex roots (conjugate);
- repeated roots.

## $n^{\text{th}}$ -Order Lin., Const. Coeff., Homogen.: Similar

Characteristic equation  $a_n r^n + \dots + a_1 r + a_0 = 0$ :

- $r$  is a root of multiplicity  $k$ :  $e^{rx}, \dots, x^{k-1}e^{rx}$ ;
- $r$  complex:  $x^l e^{rx}, x^l e^{\bar{r}x} \leftrightarrow x^l e^{ax} \cos(bx), x^l e^{ax} \sin(bx)$ .



## Example $y'' - 2y' - 3y = x$

- Undetermined Coefficients

(linear, const. coeff., special forcing)

(special forcing: poly., exp., sin or cos; their product; sum)

- $L(y) = (\frac{d}{dx} - 3)(\frac{d}{dx} + 1)y = x$

(linear, const. coeff., general forcing)

- Variation of Constant

(linear, homogeneous to inhomogeneous)

**Example**  $y'' - 2y' - 3y = \cos x + e^x$

- exp., sin or cos; sum of special forcing

**Example**  $y'' - 2y' - 3y = x^2 e^x, x \sin x, \text{ or } e^x \cos x$

- Product of poly., exp., sin or cos.

**Example**  $y'' - 2y' - 3y = e^{-x}$ ,  $y'' - 2y' + y = x^2 e^x$ .

- Need an extra factor of  $x^k$

## Linear, Const. Coeff., Inhomogeneous, Special Forcing

$$f = e^{\alpha x} [\cos(\beta x) P_1(x) + \sin(\beta x) P_2(x)]:$$

$P_1, P_2$  polynomials; max degree:  $n$ .

$\alpha + i\beta$ : soln of characteristic eq., multiplicity  $k$ .

( $P_1, P_2, k, \alpha, \beta$  can be zero.)

Try  $y_p = e^{\alpha x} [\cos(\beta x) Q_1(x) + \sin(\beta x) Q_2(x)] x^k$ :

$Q_1, Q_2$ : polynomials of degree  $\leq n$ ;

## Applications: E.g., A Spring

Spring constant  $k$ ; body of mass  $m$ ; external force  $F(t)$ ; resistive force proportional to velocity with constant  $\mu$ .

$$\text{Newton's law} \Rightarrow \frac{d^2x}{dt^2} + 2\gamma \frac{dx}{dt} + \omega_0^2 x = f(t),$$

$$\gamma = \frac{\mu}{2m} \geq 0: \text{ damping constant;}$$

$$\omega_0 = \sqrt{\frac{k}{m}} > 0: \text{ natural frequency;}$$

$$f(t) = \frac{F(t)}{m}: \text{ force (per mass).}$$

- **Energy Equation**

(not for solving, but for analyzing and interpretation)

Energy (kinetic+potential) per unit mass:

$$\frac{1}{2} \left( \frac{dx}{dt} \right)^2 + \frac{1}{2} \omega_0^2 x^2.$$

Change of energy due to damping and external forces:

$$\frac{d}{dt} \left[ \frac{1}{2} \left( \frac{dx}{dt} \right)^2 + \frac{1}{2} \omega_0^2 x^2 \right] = -2\gamma \left( \frac{dx}{dt} \right)^2 + f(t) \frac{dx}{dt}.$$

(Damping force reduces energy;

external force can reduce or increase energy.)

- **Free Oscillations** ( $f = 0$ )

$$\frac{d^2x}{dt^2} + 2\gamma\frac{dx}{dt} + \omega_0^2x = 0, \quad x(0) = a, \quad \frac{dx}{dt}(0) = 0.$$

- No damping ( $\gamma = 0$ ): simple harmonic motion  
 $x = a \cos(\omega_0 t)$ ,  $\omega_0$ : angular frequency.
- weak damping/underdamped ( $0 < \gamma < \omega_0$ ):  
decaying oscillation  $x = ae^{-\gamma t} \left[ \cos(\nu t) + \frac{\gamma}{\nu} \sin(\nu t) \right]$ ,  
frequency:  $\nu = \sqrt{\omega_0^2 - \gamma^2} < \omega_0$  natural frequency,  
amplitude: decays proportional to  $e^{-\gamma t}$ .

- critical damping ( $\gamma = \omega_0$ ): no oscillation  
 $x = a(1 + \gamma t)e^{-\gamma t}$ , tends to 0 monotonically.
- strong damping/overdamped ( $\gamma > \omega_0$ ):  
 $x = \frac{a(\lambda_2 e^{\lambda_1 t} - \lambda_1 e^{\lambda_2 t})}{\lambda_2 - \lambda_1}$ , tends to 0 monotonically.

(Another form:

$$x = ae^{-\gamma t} \left[ \cosh(\sigma t) + \frac{\gamma}{\sigma} \sinh(\sigma t) \right],$$

$$\sigma = \sqrt{\gamma^2 - \omega_0^2}.$$

Note  $\sinh(ix) = i \sin(x)$ ,  $\cosh(ix) = \cos(x)$ .)



- **Forced Oscillation and Resonance**

$$\frac{d^2x}{dt^2} + 2\gamma\frac{dx}{dt} + \omega_0^2x = f(t),$$

harmonic forcing:  $f(t) = f_0 \cos(\omega t)$ ,  $f_0 > 0$ .

$x = x_c + x_p$ ,  $x_c$  as in previous discussion.

- Damped ( $\gamma > 0$ ):  $x_c(t) \xrightarrow{t \rightarrow \infty} 0$ , transient response.

(Initial conditions to be forgotten!)

$$x_p = A \cos(\omega t - \phi), \quad A = \frac{f_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\gamma\omega)^2}}, \quad \tan\phi = \frac{2\gamma\omega}{\omega_0^2 - \omega^2}.$$

frequency: freq. of forcing;    phase shift:  $\phi \xrightarrow{\omega \uparrow \omega_0} \frac{\pi}{2}$ ;

amplitude:  $A \propto f_0$ , large if  $\omega \approx \omega_0$ ,  $\gamma \approx 0$  (resonance).

- Undamped ( $\gamma = 0$ ):  $x_c \nrightarrow 0$  (I.C. never forgotten).

- ▶  $\omega \neq \omega_0$ :  $x_p = \frac{f_0}{\omega_0^2 - \omega^2} \cos(\omega t)$ ,

$x = x_c + x_p$ : sum of oscillation of freq.  $\omega_0, \omega$ .

- ▶  $\omega = \omega_0$ :  $x_p = \frac{f_0}{2\omega_0} t \sin(\omega_0 t)$ ,

amplitude  $\rightarrow \infty$  linearly (resonance);

lag the forcing by  $\frac{\pi}{2}$ .

Energy: continuously added, no way to dissipate

$$\left( \int_0^T f \frac{dx}{dt} = \int_0^T \frac{f_0^2}{2} t \cos^2(\omega_0 t) dt + \text{bdd. osc.} \right).$$

Reality: damping or nonlinearity not included.



## Second-order Linear ODEs: General Coefficients

**Example**  $x^2y'' - 3xy' + 3y = 0$ ,  $x^2y'' - 3xy' + 4y = 0$ .

- Euler Equation: try  $y = x^r$ .

(Verify  $y = x^r \ln x$  if necessary.)

(Plug in  $x^r$ ; take  $\frac{\partial}{\partial r}$ .)

(Wronskian or Order Reduction: find  $x^r \ln x$ .)

- Try  $t = \ln x$ . Linear, const. coeff.:  $\frac{dy}{dt}$ .

## $n^{\text{th}}$ -Order Linear ODE: More

- Euler Equation  $a_n x^n y^{(n)} + \cdots + a_1 x y' + a_0 y = 0$ :

· try  $x^r$ ;      · or, let  $t = \ln x$ .

$$(y'' + p(t)y' + q(t)y = 0 \xrightarrow{x=u(t)} \text{lin. const. coeff.?)}$$

- $n^{\text{th}}$ -Order, Linear: General, Partial Results

\* Homogeneous:

- knowing 1 solution  $y_1$ : order reduction, let  $y = u y_1$   
→  $(n-1)$ -order linear homogeneous ODE of  $u'$ .  
( $n > 2$ : not often useful.)

- knowing  $n-1$  solutions: use Wronskian  
→  $(n-1)$ -order linear inhomogeneous ODE  
with known complementary solutions.

\* Inhomogeneous: variation of constant,  
complementary solutions → particular solutions.

- Series solutions
- Method of Laplace Transform

**Example**  $y'' + y = 0$ .

- Series solutions & compare.

**Example** Airy's equation  $y'' - xy = 0$ ,  $x = 0$ .

- Ordinary point  $x_0$ :  $y'' + p(x)y' + q(x)y = 0$ .  
 $p, q$ : analytic at  $x = x_0$ .
- Series solution about  $x = 1$ .

**Example** Legendre's Equation about  $x = 0$ :

$$(1 - x^2)y'' - 2xy' + l(l + 1)y = 0.$$

- $x = 0$ : ordinary point;  $x = \pm 1$ : singular points.
- Even/odd series; radius of convergence.
- $l$  integer: polynomial solutions (Legendre poly.).

## Theorem

*If  $x_0$  is an ordinary point of  $y'' + p(x)y' + q(x)y = 0$ , then the general solution of the equation is*

$$y = \sum_{n=0}^{\infty} a_n (x - x_0)^n = a_0 y_1(x) + a_1 y_2(x),$$

*where  $a_0$  and  $a_1$  are arbitrary, and  $y_1$  and  $y_2$  are two power series solutions that are analytic at  $x_0$ . The radius of convergence for  $y_1$  and  $y_2$  is at least as large as the minimum of the radii of convergence for  $p$  and  $q$ .*



**Example**  $x^4 y'' + 2x^2 y' - y = 0, x = 0.$

- Singular point: irregular.
- Series solution  $\rightarrow 0$ ;  
solutions:  $e^{\pm \frac{1}{x}}$  (essential singularity @ 0).

**Example**  $x^2 y'' + 2xy' - y = 0, x = 0.$

- Series solution  $\rightarrow 0$ ; but non-integer power!
- Singular point: regular.  $y'' + p(x)y' + q(x)y = 0$ ,  
 $x_0$ : not ordinary;  $(x - x_0)p, (x - x_0)^2 q$ : analytic at  $x_0$ .

**Example**  $x^2 y'' + xy' + (x^2 - \nu^2)y = 0$ ,  $x=0$ .  $\nu$  constant.

- 0: regular singular. Try  $y = \sum_{n=0}^{\infty} a_n x^{n+r}$ ,  $a_0 \neq 0$ .
- Indicial equation (coeff. of  $x^r$ )  $\rightarrow r_{1,2} = \pm \nu$ .
- $\nu = 0$ :  $r_1 = r_2$ , **one solution  $y_1$** .
- $\nu \neq 0$ :  $r_1 \neq r_2 \rightarrow$  two solution  $y_1, y_2$ ?
  - \*  $2\nu (= r_1 - r_2) \notin \mathbb{Z}$ :  $\checkmark$ .
  - \*  $2\nu \in \mathbb{N}$ : depends. E.g.,
    - $\nu = 1$ : **one solution  $y_1$** .
    - $\nu = \frac{1}{2}$ :  $\checkmark$  (not 3 solutions).

## Theorem

0 is a regular singular point of  $y'' + p(x)y' + q(x)y = 0$ .  $r_1$  and  $r_2$  are roots of the indicial equation.  $r_1 \geq r_2$  if real.  $\rho$  is the minimum of the radii of convergence of  $xp$  and  $x^2q$ . Then, in  $-\rho < x < 0$  or  $0 < x < \rho$ , one solution is

$$y_1(x) = |x|^{r_1} \left( 1 + \sum_{n=1}^{\infty} a_n(r_1)x^n \right).$$

For a second solution  $y_2(x)$ ,

1 If  $r_1 - r_2$  is not an integer, then

$$y_2(x) = |x|^{r_2} \left( 1 + \sum_{n=1}^{\infty} a_n(r_2)x^n \right).$$

## Theorem (Continued)

2 If  $r_1 = r_2$ , then

$$y_2(x) = y_1 \ln |x| + |x|^{r_1} \sum_{n=1}^{\infty} b_n(r_1) x^n.$$

3 If  $r_1 - r_2 \in \mathbb{N}$ , then

$$y_2(x) = ay_1 \ln |x| + |x|^{r_2} \left( 1 + \sum_{n=1}^{\infty} c_n(r_2) x^n \right),$$

where  $a$  could be zero in some exceptional case.

- $a, a_n, b_n, c_n$  can be determined by substitution. Or,
- $a_n(r)$  by recursion;  $b_n(r_1) = \frac{da}{dr}(r_1)$ ;  
 $a = \lim_{r \rightarrow r_2} (r - r_2) a_N(r)$ ,  $c_n(r) = \frac{d}{dr} [(r - r_2) a_n(r)]|_{r=r_2}$ .

**Example**  $y'' + y = \sin(2x)$ ,  $y(0) = 2$ ,  $y'(0) = 1$ .

- Laplace transform:  $\mathcal{L}\{f(x)\} = F(s) = \int_0^\infty e^{-sx} f(x) dx$ .
- $\mathcal{L}$  is linear;  $\mathcal{L}(f') = s\mathcal{L}(f) - f(0)$ ;  
 $\mathcal{L}(f * g) = \mathcal{L}(f) \cdot \mathcal{L}(g)$ ,  $f * g = \int_0^x f(x-\tau)g(\tau) d\tau$ .
- Inverse Laplace transform by table. E.g.,  
 $\mathcal{L}\{\sin(ax)\} = \frac{a}{s^2+a^2}$ ,  $\mathcal{L}\{\cos(ax)\} = \frac{s}{s^2+a^2}$ ,  $s > 0$ .

**Example**  $y'' + y' - 2y = g(x)$ ,  $y(0) = y'(0) = 0$ ,

$$g(x) = \begin{cases} 1, & 5 \leq x < 20, \\ 0, & 0 \leq x < 5 \text{ or } x \geq 20. \end{cases}$$

- Piecewise forcing.
- Solve over 3 intervals separately, 6 free constants;  
I.C. + continuity of  $y, y'$  at 5, 20  $\rightarrow$  constants.
- $g(x) = u_5(x) - u_{20}(x)$ , apply Laplace transform.

$$u_c(x) = \begin{cases} 1, & x \geq c, \\ 0, & x < c. \end{cases} \quad \begin{aligned} \mathcal{L}\{u_c(x)f(x-c)\} &= e^{-cs}\mathcal{L}(f), \\ \mathcal{L}\{e^{cx}f(t)\} &= \mathcal{L}(f)(s-c) \end{aligned}$$

**Example**  $y'' + y = 2\delta(x - 5)$ ,  $y(0) = y'(0) = 0$ .

- Impulse forcing: Dirac delta function. (Next slide!)
- Solve for  $0 < x < 5$  and  $x > 5$ : 4 constants.  
I.C.,  $y(5-) = y(5+)$ ,  $y'(5+) - y'(5-) = 2 \rightarrow \text{const.}$
- Apply Laplace transform.
- Note: impulse forcing  $\leftrightarrow$  adding a velocity

## Dirac Delta Function

$$\delta(x) = \begin{cases} \infty, & x = 0, \\ 0, & x \neq 0. \end{cases}$$

- $\delta(x)$ : a distribution (generalized function).

Defining property:  $\int_{-\infty}^{\infty} f(x)\delta(x)dx = f(0), \forall f$  continuous.

- $\delta(x) = u'_0(x)$  in distributional sense.

- $\delta(x)$ : 'limit' of  $\delta_\epsilon(x)$  as  $\epsilon \rightarrow 0$ .

$$\delta_\epsilon(x) = \frac{\epsilon}{\pi(x^2 + \epsilon^2)}, \frac{1}{\sqrt{2\pi}\epsilon^2} e^{-x^2/(2\epsilon^2)}, \text{ or } \frac{1}{2\epsilon} \mathbf{1}_{\{-\epsilon < x < \epsilon\}}, \text{ etc.}$$

(unit area & vanishing width.)

- 'density' of point mass, point charge; force in collision.



# Higher Order ODEs: General

- Order Reduction via integration, substitution, etc: e.g.,
  - $y'' = f(x, y')$ : let  $u = y'$ , then  $u' = f(x, u)$ .
  - $y'' = f(y, y')$ : let  $u = y'$  and  $y$  independent variable, then,  $u \frac{du}{dy} = f(y, u)$ .
- . . . . .

# System of ODEs

## (First-Order Linear)

**Example** 
$$\begin{cases} \frac{dy}{dx} = 3y - 2z , \\ \frac{dz}{dx} = 2y - z . \end{cases}$$

- Elimination Method.
- De-coupling Method.
- Try exponential form.
- Solution in Matrix Form.

**Example** 
$$\begin{cases} \frac{d^2x}{dt^2} + \frac{dy}{dt} - x = e^t, \\ \frac{d^2y}{dt^2} + \frac{dx}{dt} + y = 0. \end{cases}$$

- Elimination Method.
- Higher  $\rightarrow$  1<sup>st</sup>-Order: more variables, more functions.
- Linear inhomogeneous system (undetermined coefficient; variation of constant; Laplace transform)